

Lesson 1

Python Basics

Code the Dream — Intro to Python

EXPLORE

Today's Session

EXPLORE

- Variables & data types
- `print()` and `input()`
- Type conversion
- Operators & f-strings
- Scripts & debugging

APPLY

- Profile Card Builder
- Code-along
- Extension challenge

By the end: store data in variables, collect user input, display formatted results, and read Python error messages.

EXPLORE

Variables

A variable is a **named container** that stores a value.

```
name = "Jazmine"  
age  = 28  
height = 5.8
```

= means **assign** — not "equals" like in math. "Put the value on the right into the name on the left."

EXPLORE

The Four Core Data Types

Type	Keyword	Example
Whole number	int	42
Decimal number	float	3.14
Text	str	"hello"
True / False	bool	True

```
print(type(42))      # <class 'int'>
print(type("hello")) # <class 'str'>
print(type(True))    # <class 'bool'>
```

EXPLORE

print() and input()

print() — show output

```
name = "Jazmine"  
print("Hello,", name)  
# Hello, Jazmine
```

input() — ask for input

```
name = input("Your name? ")  
print("Hello,", name)
```



input() **always returns a string** — even if the user types a number.

Check for Understanding

What will this code output?

```
age = input("How old are you? ")  
print(type(age))
```

- A) `<class 'int'>` — if the user typed a number
- B) `<class 'str'>` — always
- C) It depends on what the user types
- D) An error

EXPLORE

Type Conversion

`input()` always returns a string. Convert before doing math.

```
# CRASHES – can't add a string and an integer
age = input("Your age? ")
next_year = age + 1          # TypeError!
```

```
# Works – convert right where you collect input
age = int(input("Your age? "))
next_year = age + 1
print(f"Next year you'll be {next_year}")
```

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Operators

`//` — integer division (drops the decimal)

```
7 // 2    # 3    (not 3.5)
```

`%` — remainder after division

```
7 % 3    # 1    (7 = 2×3 + 1)
```

Comparison operators return `True` or `False`:

```
age = 20
print(age > 18)    # True
print(age == 25)   # False
```

`==` checks equality. `=` assigns a value.

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f-strings

Add `f` before the opening quote, then use `{ }` around any variable or expression.

```
name = "Jazmine"  
age  = 28  
print(f"My name is {name} and I am {age} years old.")  
# My name is Jazmine and I am 28 years old.
```

```
total = 0.1 + 0.2  
print(f"{total:.2f}")    # 0.30 – controls decimal places
```

EXPLORE

How Scripts Run

Python reads your file **top to bottom, one line at a time**.

```
print(greeting)          # NameError – 'greeting' doesn't exist yet  
greeting = "Hello!"
```

Build → Test → Build

Write a few lines, run it, confirm it works, then write more. Don't wait until the end to run your code.

Check for Understanding

What error will this code produce, and why?

```
Name = "Jazmine"  
print(f"Hello, {name}!")
```

- A) `SyntaxError` — missing parenthesis
- B) `TypeError` — wrong data type
- C) `NameError` — `name` is not defined
- D) No error — it prints normally

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Reading Error Messages

Read from the bottom up. The last line tells you the most useful information.

```
File "main.py", line 4
    next_year = age + 1
TypeError: can only concatenate str (not "int") to str
```

1. **Where:** `line 4` — which line caused the problem
2. **What:** `TypeError` — the category of error
3. **Why:** `can only concatenate str...` — what went wrong

EXPLORE

Five Errors to Know

Error	What it usually means
<code>SyntaxError</code>	Python can't read the code — missing quote, paren, or colon
<code>NameError</code>	Used a name Python doesn't recognize — typo or wrong order
<code>TypeError</code>	Wrong type for the operation — e.g., string + integer
<code>ValueError</code>	Right type, nonsense value — e.g., <code>int("hello")</code>
<code>IndentationError</code>	Unexpected spaces at the start of a line

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Debugging with `print()`

When code runs but gives the wrong output, add temporary prints to inspect values.

```
bill = input("Bill amount? $")
print("DEBUG bill:", bill, type(bill))

bill = float(bill)
print("DEBUG after conversion:", bill)

tip = bill * 0.20
print("DEBUG tip:", tip)
```

Check for Understanding

What will this code produce?

```
age = "25"  
print(age + 5)
```

- A) 30
- B) "255"
- C) A `TypeError`
- D) A `ValueError`

Now: Apply

Time to write some Python.

APPLY

This Week's Assignment: Profile Card Builder

One script, built up in five sections:

1. Variables and types
2. User input and math
3. Type conversion and f-strings
4. Formatted receipt
5. Mini-project: full profile card

Open **online-python.com** — we'll build sections 2, 3, and 4 together.

APPLY

Code-Along: User Input + Math

```
name = input("What is your name? ")
birth_year = int(input("What year were you born? "))

age = 2026 - birth_year

print(f"Hi, {name}! You are approximately {age} years old.")
```

Why `int()` ? Because `input()` returns a string — you need a number to do math.

APPLY

Code-Along: Two Numbers, One Result

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

result = num1 * num2

print(f"{num1} × {num2} = {result:.2f}")
```

Try it: Enter `12.5` and `4` . What do you see?

APPLY

Code-Along: Formatted Receipt

```
item = "Python textbook"
price = 29.99
qty = 2
total = price * qty

print("=====")
print("          RECEIPT")
print("=====")
print(f"Item:      {item}")
print(f"Price:      ${price:.2f}")
print(f"Quantity:   {qty}")
print("-----")
print(f"Total:      ${total:.2f}")
print("=====")
```

APPLY

Extension: Full Profile Card

If you finish early, try Section 5 — the full profile card.

PROFILE: Alex Rivera

Hometown: Chicago, IL
Hobby: Rock climbing
Fun fact: I've visited 12 countries.
Age: 27

Challenge: Compute age from birth year — don't ask for it directly. How would you align the labels?

EXPLORE

Wrap-Up

This week you learned to:

- Store values in variables with the right data type
- Collect user input and convert it for math
- Display formatted output with f-strings
- Read error messages and debug with `print()`

Before next session: Complete the assignment on online-python.com. Submit your share link + video reflection in CTD Learns. Post questions in **#[class-name]-discussion** on Slack.